



Representation

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Principal  
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Description

Principal Components

# Chapter 8.1

## Representation and Description

*Image Processing and Computer Vision*

**LE Thanh Sach**

*Faculty of Computer Science and Engineering  
Ho Chi Minh University of Technology, VNU-HCM*

# Overview

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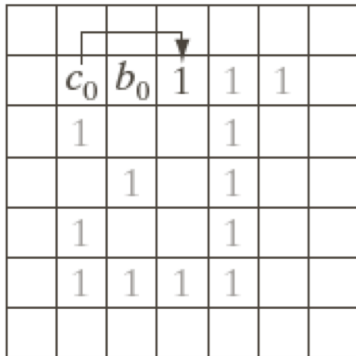
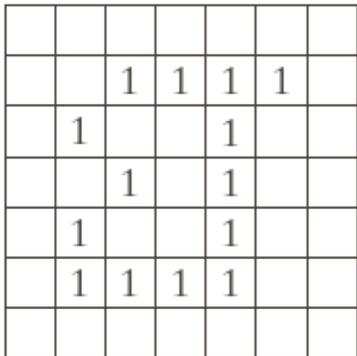
- Principal Components

# Representation: Boundary Following

## Algorithm:

### Step 1: Initialization

- $b_0$ : The uppermost and leftmost point belonging to the region
- $c_0$ : The west neighbor of  $b_0$



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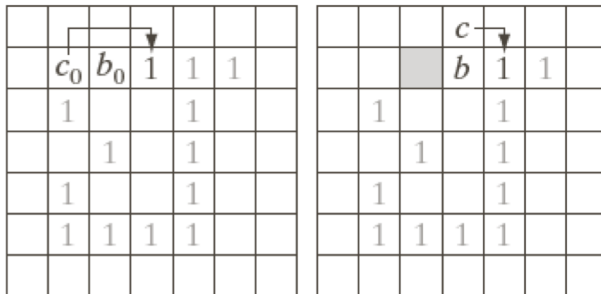
Principal Components

## Representation: Boundary Following

### Algorithm:

#### Step 2: Finding next point

- Let the 8-neighbors of  $b_0$ , starting at  $c_0$  and proceeding in the clockwise direction be denoted as  $n_1, n_2, \dots, n_8$
- Find the first  $n_k$  in this sequence such that  $n_k = 1$
- Update  $b_1 = n_k$  and  $c_1 = n_{k-1}$



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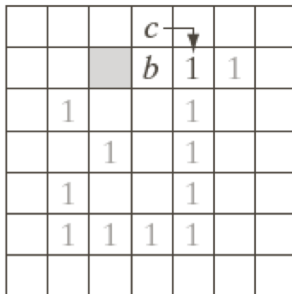
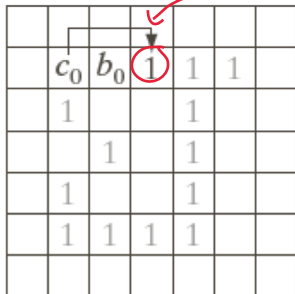
Principal Components

## Representation: Boundary Following

### Algorithm:

#### Step 3: Finding next point

- Let  $b = b_1$  and  $c = c_1$
- Let the 8-neighbors of  $b$ , starting at  $c$  and proceeding in the clockwise direction be denoted as  $n_1, n_2, \dots, n_8$
- Find the first  $n_k$  in this sequence such that  $n_k = 1$
- Update  $b = n_k$  and  $c = n_{k-1}$



# Representation: Boundary Following

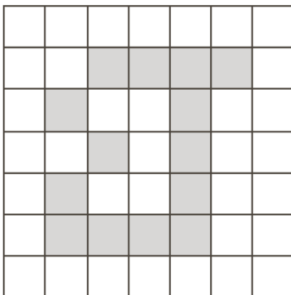
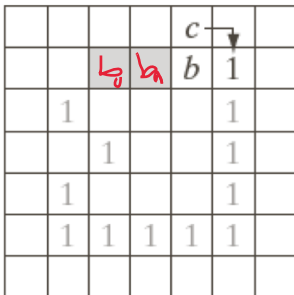
## Algorithm:

### Step 4: Iteration

- Repeat **Step 3** until

①  $b = b_0$  **AND**

② The next boundary point of  $b$  is  $b_1$

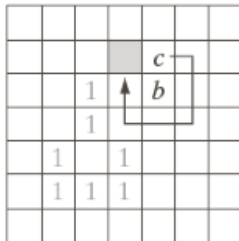
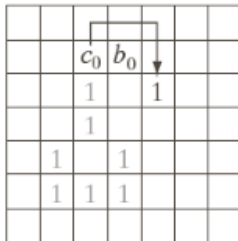
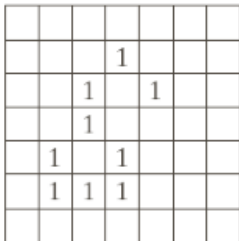


# Representation: Boundary Following

## Algorithm:

### Questions

- 1 How does the algorithm work, if the stopping condition is only  $b = b_0$ , as shown in the following figure?
- 2 How can we detect the boundary of holes, i.e., zero-values surrounded with 1-values.

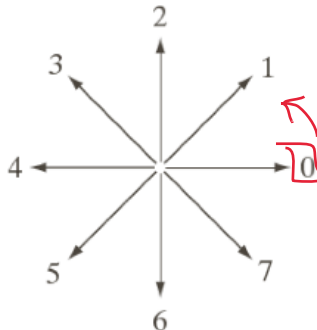
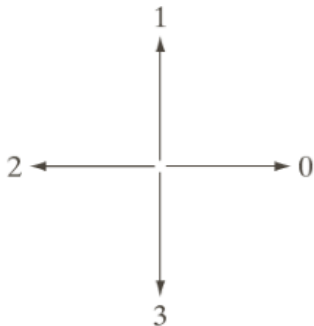


## Representation: Chain codes

### Definition

**Feeman chain codes**  $\equiv$  **A sequence of straight-line segments** of specified length and direction.

Each segment, as shown below, is denoted a number according 4- or 8-connectivity:





### Algorithm for extracting chain codes

- ① Re-sample the boundary by using a larger grid spacing
- ② For each boundary point (during boundary traversing)
  - ① Obtain the node in the larger grid that is nearest to the current boundary point.
  - ② If this node is different to the node in previous step,
    - Compute a code using 4- or 8-connectivity from the node in the previous step to the current node
    - Output this code.

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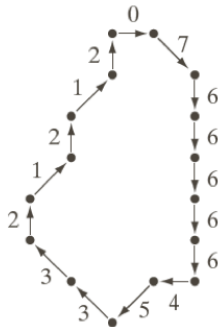
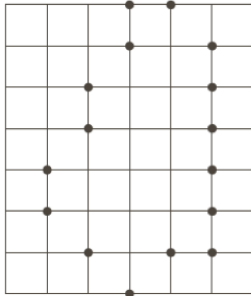
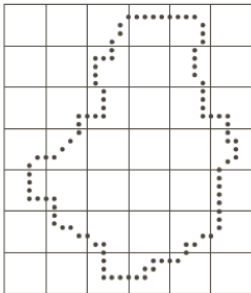
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# Representation: Chain codes

## A example



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# Representation: Chain codes

## Problems facing with chain codes

The chain code of a boundary depends on the starting point.

### Solutions:

- ① Sequence of codes forming the minimum integer
  - Consider the chain code as a circular sequence.
  - Re-define the starting point so that the sequence forms an integer with minimum magnitude.
- ② Sequence of differences
  - Difference  $\equiv$  number of direction changes (e.g., in counterclockwise) between two adjacent codes.



# Representation: Chain codes

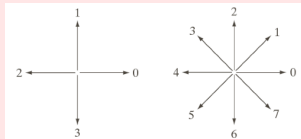
## Problems facing with chain codes

The chain code of a boundary depends on the starting point.

### Solutions:

#### ② Sequence of differences

- Difference  $\equiv$  number of direction changes (e.g., in counterclockwise) between two adjacent codes. For 4-connectivity:



- From 1 to 0  $\equiv$  3
- From 1 to 3  $\equiv$  2
- From 2 to 0  $\equiv$  2



# Representation: Chain codes

## Problems facing with chain codes

The chain code of a boundary depends on the starting point.

### Solutions:

- ② Sequence of differences
- ③ Represent the chain code by a sequence of differences instead of the chain code itself.
- ④ The first difference  $\equiv$  the difference between the last and the first components in the original. For 4-connectivity:
  - 10103322  $\rightarrow$  33133030



## Representation: Chain codes



### Questions

- 1 How can you extract the outer and the inner boundary of the ring in the figure? Note: there are noise in the input image.
- 2 How to extract the chain code that is invariant to rotation or orientation?



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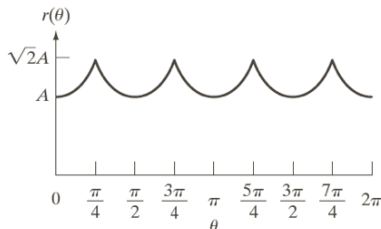
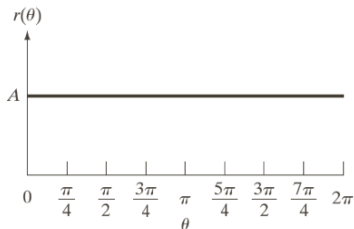
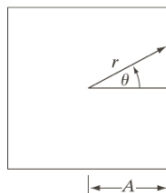
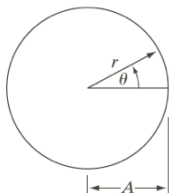
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# Representation: Signature



## Ideas

Convert 2D boundary to 1D time series data.



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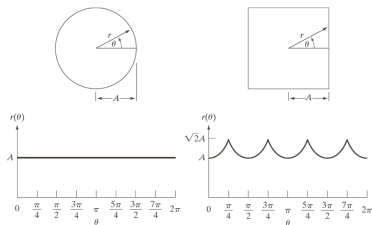
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# Representation: Signature



## A Method: Distance from the centroid

- ① Determine the centroid of the boundary
- ② For each angle  $\theta$  (discretized angle)
  - Compute the distance from the centroid to the boundary point on the line oriented angle  $\theta$
  - Output this distance



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# Representation: Signature

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### Problem facing with signature

- ① Signatures depend on the starting point, not invariant to orientation (rotation)

### Solutions:

- ① Starting point  $\equiv$  The farthest point from the centroid (assume that it is unique)
- ② Starting point  $\equiv$  The farthest point on eigen-axis

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# Representation: Signature

## Problem facing with signature

- ② The magnitude of signatures depends on image scaling

## Solutions:

- ① Normalize the magnitude to a certain range, for examples,  $[0, 1]$



# Representation: Signature

## Another Method: Histogram of angles

Input: **A reference line**  $L_{ref}$

- ① Extract boundary points
- ② For each point  $p$  on the boundary
  - Compute tangent line  $L_{tangent}$  to the boundary at  $p$
  - Compute the angle  $\theta$  between  $L_{tangent}$  and  $L_{ref}$
  - Output  $\theta$

- $L_{ref} \equiv Ox \Rightarrow$  The method
- $\equiv$  Histogram of tangent angle values
- $\equiv$  Slope-density function
- $\equiv$  Histogram of orientation for boundary points (HOG)



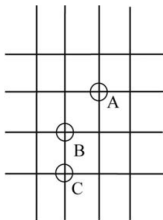
# Basic Boundary Descriptors

## Perimeter

Perimeter is the length of boundary.

## Approximation methods:

- ①  $\equiv$  The number of pixels on the boundary.
- ②  $\equiv N_e + N_o\sqrt{2}$ 
  - $N_e$  : Number of even codes in chain code representation
  - $N_o$  : Number of odd codes in chain code representation



- $\text{distance}(B,C) = 1$ ; but,  $\text{distance}(A,B) = \sqrt{2}$





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## Diameter

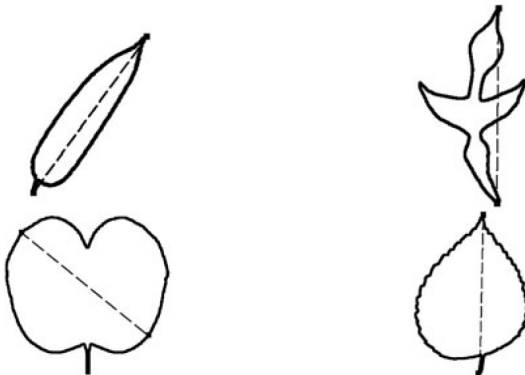
Diameter is the largest distance between any two points on a boundary.

## Mathematical Definition:

- $Diam(B)$ : Diameter of boundary  $B$
- $D(p_i, p_j)$ : The distance between two points  $p_i$  and  $p_j$  on the boundary  $B$

$$Diam(B) = \max_{(i,j)} \{D(p_i, p_j)\}$$

# Basic Boundary Descriptors



**Figure:** Diameter demonstration (Source: Book on Shape Classification and Analysis)



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## Major and Minor Axes

- **Major Axis** : The direction along which the shape points are more dispersed. The distance between two farthest points on the major axis is the **diameter**.
- **Minor Axis** : The direction that is perpendicular to the major axis

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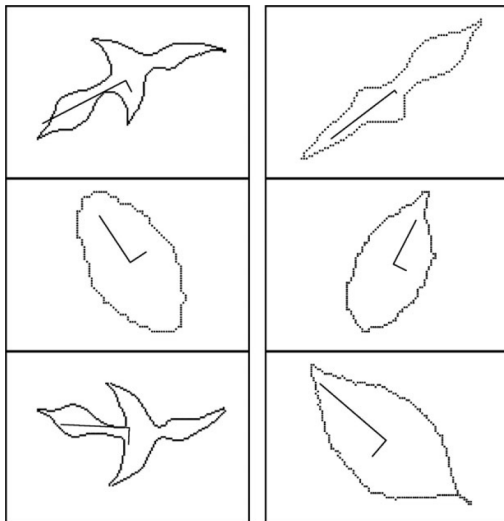
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**Figure:** Demonstration of Major and Minor axes (Source: Book on Shape Classification and Analysis)



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### Algorithm for obtaining major and minor axes

- ① Extract boundary points, using boundary following algorithm. Assume that there are  $n$  boundary points
- ② Store  $n$  boundary points into matrix  $X$  of size  $n \times 2$ 
  - Each boundary point  $p(x_p, y_p)$  is stored into a row in  $X$
- ③ Compute covariance matrix  $K$  of  $X$ .
- ④ Calculate the eigenvectors and eigenvalues of  $K$ 
  - Major axis is the eigenvector corresponding to the largest eigenvalue.
  - Minor axis is the remaining eigenvector.



## Basic rectangle and Eccentricity

- **Basic rectangle** : The rectangle is a smallest rectangle that is aligned with the major and the minor axes and completely encloses the boundary
- **Eccentricity** :

$$E = \frac{\text{length}(\text{major axis})}{\text{length}(\text{minor axis})}$$

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## Shape number

- **Shape number** : The first difference (see chain-code) of smallest magnitude
- **Order of shape** : The order of shape  $\equiv$  The number of digits in the shape number

# Basic Boundary Descriptors

Order 4



Chain code: 0 3 2 1

Difference: 3 3 3 3

Shape no.: 3 3 3 3

Order 6

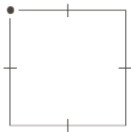


Chain code: 0 0 3 2 2 1

Difference: 3 0 3 3 0 3

Shape no.: 0 3 3 0 3 3

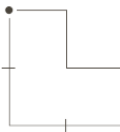
Order 8



Chain code: 0 0 3 3 2 2 1 1

Difference: 3 0 3 0 3 0 3 0

Shape no.: 0 3 0 3 0 3 0 3



Chain code: 0 3 0 3 2 2 1 1

Difference: 3 3 1 3 3 0 3 0

Shape no.: 0 3 0 3 3 1 3 3



Chain code: 0 0 0 3 2 2 2 1

Difference: 3 0 0 3 3 0 0 3

Shape no.: 0 0 3 3 0 0 3 3

Dot is the starting point



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# Basic Boundary Descriptors

## Questions

Given a order, How do we obtain encode a shape in this order?

## Solutions

- 1 Determine the basic rectangle of the shape.
- 2 Calculate the eccentricity of the basic rectangle, called  $E_1$
- 3 From the input order, find a rectangle, called  $R$ , that has the order and best approximates  $E_1$ .
  - Input order  $n = 12$
  - $\Rightarrow n = 1 \times 12$  ( $E = 12$ );  $n = 2 \times 6$  ( $E = 3$ );  $n = 3 \times 4$  ( $E = \frac{4}{3}$ ), etc.
- 4 Use  $R$  to create the grid size.
- 5 Use this grid to generate first difference and shape number.



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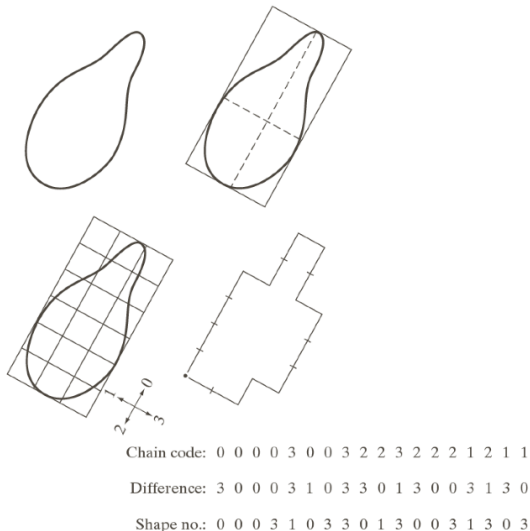
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**Figure:** Demonstration for generating shape number from an order



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# Fourier Descriptors

## Input

- A shape

## Method

- ① Extract boundary points:  $(x_0, y_0), (x_1, y_1)$ , etc
- ② Define a sequence of complex numbers:

$$s(k) = x(k) + jy(k)$$

- $x(k) = x_k; y(k) = y_k$
- $k = 0, 1, \dots, N - 1$



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# Fourier Descriptors

## Method

### ③ Perform Discrete Fourier Transform (DFT)

$$a(u) = \sum_{k=0}^{N-1} s(k) e^{-j2\pi uk/N}$$

- $u = 0, 1, \dots, N - 1$
- $a(u)$  are **Fourier Descriptors**

## Inverse DFT

$s(k)$  can be approximated from the first  $P$  Fourier Descriptors as follows:

$$\hat{s}(k) = \frac{1}{P} \sum_{u=0}^{P-1} a(u) e^{j2\pi uk/N}$$



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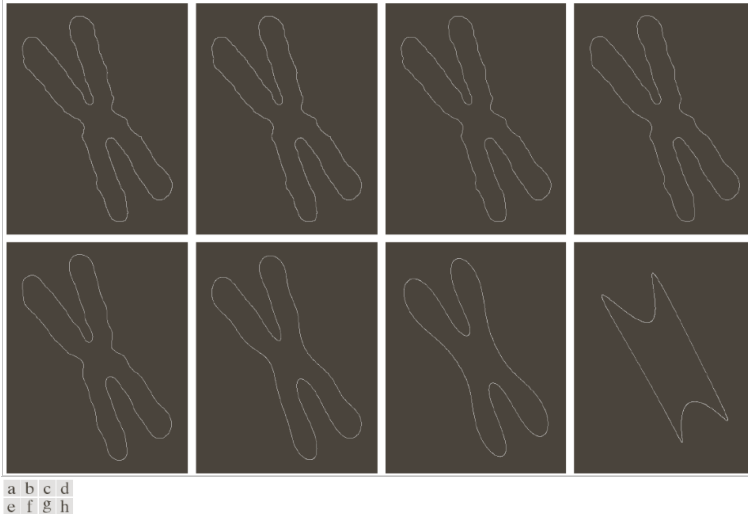
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# Fourier Descriptors: Example



**FIGURE 11.20** (a) Boundary of human chromosome (2868 points). (b)–(h) Boundaries reconstructed using 1434, 286, 144, 72, 36, 18, and 8 Fourier descriptors, respectively. These numbers are approximately 50%, 10%, 5%, 2.5%, 1.25%, 0.63%, and 0.28% of 2868, respectively.



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| Transformation | Boundary                      | Fourier Descriptor                     |
|----------------|-------------------------------|----------------------------------------|
| Identity       | $s(k)$                        | $a(u)$                                 |
| Rotation       | $s_r(k) = s(k)e^{j\theta}$    | $a_r(u) = a(u)e^{j\theta}$             |
| Translation    | $s_t(k) = s(k) + \Delta_{xy}$ | $a_t(u) = a(u) + \Delta_{xy}\delta(u)$ |
| Scaling        | $s_s(k) = \alpha s(k)$        | $a_s(u) = \alpha a(u)$                 |
| Starting point | $s_p(k) = s(k - k_0)$         | $a_p(u) = a(u)e^{-j2\pi k_0 u/K}$      |

**TABLE 11.1**  
Some basic  
properties of  
Fourier  
descriptors.

## Regional Simple Descriptors

- **Perimeter:** defined in previous section
- **Area:** The number of pixels in the shape
- **Compactness:**

$$\text{Compactness} = \frac{\text{Perimeter}^2}{\text{Area}}$$

- **Circularity Ratio:** The ratio of the area of the input region to the area of the circle that has the same perimeter with the input region

$$R_c = \frac{4\pi A}{P^2}$$

- $A$ : Area of the region.
- $P$ : Perimeter of the region.
- $R_c = 1 \Rightarrow$  Circle;  $R_c = \pi/4 \Rightarrow$  Square;



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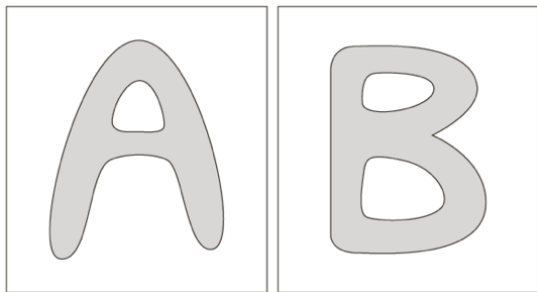
### Moments

### Principal Components for Description

Principal Components

# Topological Descriptors

- **Input:** A region has  $H$  holes and  $C$  connected components
- **Euler number:**  $E = C - H$



a b

**FIGURE 11.25**  
Regions with Euler numbers equal to 0 and  $-1$ , respectively.



## Representation

Boundary following  
Chain codes  
Signatures

## Boundary Descriptors

Basic Descriptors  
Fourier Descriptors

## Regional Descriptors

Simple Descriptors  
**Topological Descriptors**  
Texture  
Spectral Descriptors

## Moments

## Principal Components for Description

Principal Components

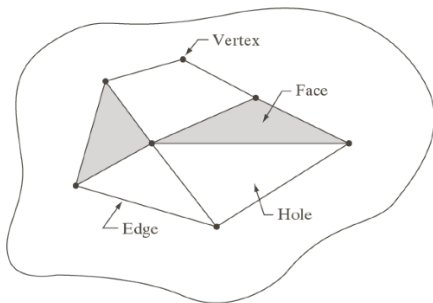
# Topological Descriptors

- **Input:** A Region represented by straight-line segments

- ①  $F$  : Number of faces
- ②  $V$  : Number of vertices
- ③  $Q$  : Number of edges

- **Euler number:**

$$\begin{aligned} E &= C - H \\ &= V - Q + F \end{aligned}$$



**FIGURE 11.26** A region containing a polygonal network.



## Representation

Boundary following

Chain codes

Signatures

## Boundary Descriptors

Basic Descriptors

Fourier Descriptors

## Regional Descriptors

Simple Descriptors

Topological Descriptors

Texture

Spectral Descriptors

## Moments

## Principal Components for Description

Principal Components

## Histogram-based descriptors:

### Input:

- An image or a region

### A Method

#### ① Compute histogram

$$p(z) = [p(z_0), p(z_1), p(z_i) \dots, p(z_{L-1})]$$

- $i = 0, 1, \dots, L - 1$  (  $L = 256$  for gray image)

#### ② Compute the mean of $z$

$$m = \sum_{i=0}^{L-1} z_i p(z_i)$$



### Representation

Boundary following  
Chain codes  
Signatures

### Boundary Descriptors

Basic Descriptors  
Fourier Descriptors

### Regional Descriptors

Simple Descriptors  
Topological Descriptors

### Texture

Spectral Descriptors

### Moments

### Principal Components for Description

Principal Components



## Histogram-based descriptors:

### A Method

- ③ Compute the  $n$ th moments about the mean  $m$

$$\mu_n = \sum_{i=0}^{L-1} (z_i - m)^n p(z_i)$$

#### Representation

Boundary following  
Chain codes  
Signatures

#### Boundary Descriptors

Basic Descriptors  
Fourier Descriptors

#### Regional Descriptors

Simple Descriptors  
Topological Descriptors

#### Texture

Spectral Descriptors

#### Moments

#### Principal Components for Description

Principal Components

### Histogram-based descriptors:

#### Moments' meaning and properties

- ①  $n = 0$ :  $\mu_0 = 1$
- ②  $n = 1$ :  $\mu_1 = 0$
- ③  $n = 2$ :  $\mu_2 = \sigma^2$  : measure the variance of  $z$  (intensities)
- ④ **Relative Smoothness**  $R$  is defined as

$$R = 1 - \frac{1}{1 + \sigma^2}$$

- $R = 0$ : for region of constant intensity
- $R \rightarrow 1$  (approaching 1) for large variance  $\sigma^2$
- $\sigma^2$  should be normalized to 1 by  $\sigma^2 \leftarrow \sigma^2 / (L - 1)^2$



#### Representation

Boundary following  
Chain codes  
Signatures

#### Boundary Descriptors

Basic Descriptors  
Fourier Descriptors

#### Regional Descriptors

Simple Descriptors  
Topological Descriptors

#### Texture

Spectral Descriptors

#### Moments

#### Principal Components for Description

Principal Components



## Histogram-based descriptors:

### Moments' meaning and properties

- ⑤  $n = 3$ :  $\mu_3$  measures the skewness of the region's histogram.
- ⑥  $n = 4$ :  $\mu_4$  measures the flatness of the region's histogram.
- ⑦  $n \geq 5$ : still provide further **discriminative** information of texture content.

#### Representation

Boundary following  
Chain codes  
Signatures

#### Boundary Descriptors

Basic Descriptors  
Fourier Descriptors

#### Regional Descriptors

Simple Descriptors  
Topological Descriptors

#### Texture

Spectral Descriptors

#### Moments

#### Principal Components for Description

Principal Components



## Representation

Boundary following  
Chain codes  
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## Boundary Descriptors

Basic Descriptors  
Fourier Descriptors

## Regional Descriptors

Simple Descriptors  
Topological Descriptors

## Texture

Spectral Descriptors

## Moments

## Principal Components for Description

Principal Components

## Histogram-based descriptors:

### More descriptors using histogram

#### Uniformity $U$ :

$$U = \sum_{i=0}^{L-1} p(z_i)^2$$

- $U$  is maximum for images in which all intensities are equal.



## Representation

Boundary following  
Chain codes  
Signatures

## Boundary Descriptors

Basic Descriptors  
Fourier Descriptors

## Regional Descriptors

Simple Descriptors  
Topological Descriptors

## Texture

Spectral Descriptors

## Moments

## Principal Components for Description

Principal Components

## Histogram-based descriptors:

### More descriptors using histogram

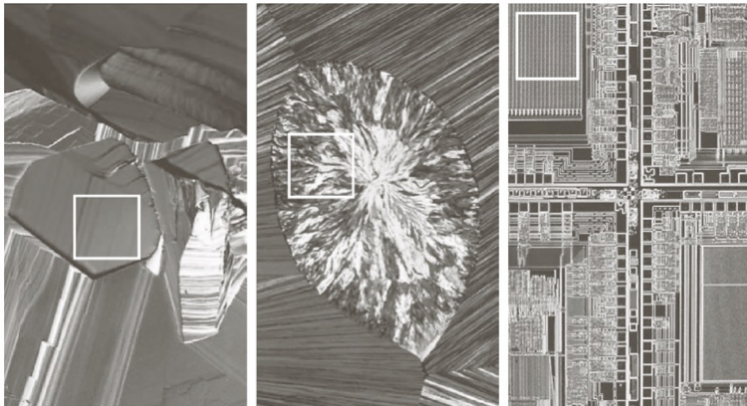
#### Average Entropy $E$ :

$$E = - \sum_{i=0}^{L-1} p(z_i) \log_2 p(z_i)$$

- $E$  measures the variation of intensities in images.  
 $E = 0$  for constant images.

# Texture: Statistical Approaches

## Histogram-based descriptors:



**Figure:** Three kinds of texture in sub-images



### Representation

- Boundary following
- Chain codes
- Signatures

### Boundary Descriptors

- Basic Descriptors
- Fourier Descriptors

### Regional Descriptors

- Simple Descriptors
- Topological Descriptors

### Texture

- Spectral Descriptors

### Moments

### Principal Components for Description

- Principal Components



## Histogram-based descriptors:

| Texture | Mean   | Standard deviation | $R$ (normalized) | Third moment | Uniformity | Entropy |
|---------|--------|--------------------|------------------|--------------|------------|---------|
| Smooth  | 82.64  | 11.79              | 0.002            | -0.105       | 0.026      | 5.434   |
| Coarse  | 143.56 | 74.63              | 0.079            | -0.151       | 0.005      | 7.783   |
| Regular | 99.72  | 33.73              | 0.017            | 0.750        | 0.013      | 6.674   |

**Figure:** Measurements for the previous three kinds of textures

Find the **discrimination** on the above measurements

### Representation

Boundary following  
Chain codes  
Signatures

### Boundary Descriptors

Basic Descriptors  
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### Regional Descriptors

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### Principal Components for Description

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## Representation

Boundary following  
Chain codes  
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## Boundary Descriptors

Basic Descriptors  
Fourier Descriptors

## Regional Descriptors

Simple Descriptors  
Topological Descriptors

## Texture

Spectral Descriptors

## Moments

## Principal Components for Description

Principal Components

## Relative positions of pixels:

### Input:

- An input image or a region
- $Q$  an operator that defines the position of two pixels relative to each other.
- $L$ : number of intensity levels.
- Intensities are transformed to range of  $[1, L]$  instead of  $[0, L - 1]$



## Representation

Boundary following  
Chain codes  
Signatures

## Boundary Descriptors

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## Regional Descriptors

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## Moments

## Principal Components for Description

Principal Components

## Relative positions of pixels:

### Concept

**Co-occurrence matrix:**  $G = \{g_{ij}\}_{L \times L}$

- $G$  has size of  $L \times L$
- $i, j \in [1, L]$

**Meaning of  $g_{ij}$ :**  $g_{ij}$  is the number of pairs of two pixels  $p$  and  $q$  that satisfy the following predicates

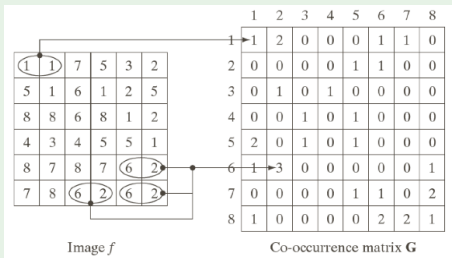
- $p$  and  $q$  satisfy operator  $Q$ , for examples, they are two consecutive pixels in the input image
- Gray level of pixel  $p$  is  $i$
- Gray level of pixel  $q$  is  $j$

# Texture: Statistical Approaches

## Relative positions of pixels:

### Examples

- $Q$  : one pixel immediately to the right
- $L = 8$
- Input image  $f$  of small size,  $6 \times 6$



**Figure:** Demonstration of building co-occurrence matrix



### Representation

Boundary following

Chain codes

Signatures

### Boundary Descriptors

Basic Descriptors

Fourier Descriptors

### Regional Descriptors

Simple Descriptors

Topological Descriptors

### Texture

Spectral Descriptors

### Moments

### Principal Components for Description

Principal Components

## Texture: Statistical Approaches

### Relative positions of pixels:

#### Normalized Co-occurrence Matrix: $G_{norm}$

$$G_{norm} = \{p_{ij} = \frac{g_{ij}}{n}\}_{L \times L}$$

- $n$  : the total number of pairs that satisfy  $Q$

$$n = \sum_{i=1}^L \sum_{j=1}^L g_{ij}$$

- $p_{ij}$  : probability of occurring a pair of two pixels  $p$  and  $q$  that satisfy the positional operator  $Q$  and that they have gray-levels  $i$  and  $j$  respectively.



#### Representation

Boundary following  
Chain codes  
Signatures

#### Boundary Descriptors

Basic Descriptors  
Fourier Descriptors

#### Regional Descriptors

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#### Texture

Spectral Descriptors

#### Moments

#### Principal Components for Description

Principal Components

## Relative positions of pixels:

### Descriptors bases on $G_{norm}$

#### ① Maximum Probability :

$$\max_{(i,j)} \{p_{ij}\}$$

- Measures the strongest response of the co-occurrence



#### Representation

Boundary following  
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## Relative positions of pixels:

Descriptors bases on  $G_{norm}$

### ② Correlation :

$$\sum_{i=1}^K \sum_{j=1}^K \frac{(i - m_r)(j - m_c)p_{ij}}{\sigma_r \sigma_c}$$

- Measures how correlated a pixel is to its neighbors over the entire image
- Range of values:  $[-1, 1]$ , i.e., perfect negative and perfect positive.
- $\sigma_r \neq 0$  and  $\sigma_c \neq 0$



#### Representation

Boundary following  
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# Texture: Statistical Approaches

## Relative positions of pixels:

### Descriptors bases on $G_{norm}$

#### ② Correlation :

$$m_r = \sum_{i=1}^K i \sum_{j=1}^K p_{ij}$$

$$m_c = \sum_{j=1}^K j \sum_{i=1}^K p_{ij}$$

$$\sigma_r = \sum_{i=1}^K (i - m_r)^2 \sum_{j=1}^K p_{ij}$$

$$\sigma_c = \sum_{j=1}^K (j - m_c)^2 \sum_{i=1}^K p_{ij}$$



#### Representation

Boundary following  
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Signatures

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## Relative positions of pixels:

### Descriptors bases on $G_{norm}$

#### ③ Contrast :

$$\sum_{i=1}^K \sum_{j=1}^K (i - j)^2 p_{ij}$$

- Measures the intensity contrast between a pixel to its neighbors over the entire image.
- Range of values:  $[0, (K - 1)^2]$
- Contrast = 0  $\equiv G$  is constant.



#### Representation

Boundary following  
Chain codes  
Signatures

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Fourier Descriptors

#### Regional Descriptors

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#### Moments

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Principal Components

## Relative positions of pixels:

### Descriptors bases on $G_{norm}$

#### ④ Uniformity :

$$\sum_{i=1}^K \sum_{j=1}^K p_{ij}^2$$

- Measures the uniformity
- Range of values:  $[0, 1]$
- Uniformity = 1 for constant images



### Representation

Boundary following  
Chain codes  
Signatures

### Boundary Descriptors

Basic Descriptors  
Fourier Descriptors

### Regional Descriptors

Simple Descriptors  
Topological Descriptors

### Texture

Spectral Descriptors

### Moments

### Principal Components for Description

Principal Components

## Relative positions of pixels:

### Descriptors bases on $G_{norm}$

#### ⑤ Homogeneity :

$$\sum_{i=1}^K \sum_{j=1}^K \frac{p_{ij}}{1 + |i - j|}$$

- Measures the spatial closeness of the distribution of elements in  $G$  to the diagonal
- Range of values:  $[0, 1]$
- Homogeneity = 1  $\equiv G$  is diagonal matrix



#### Representation

Boundary following  
Chain codes  
Signatures

#### Boundary Descriptors

Basic Descriptors  
Fourier Descriptors

#### Regional Descriptors

Simple Descriptors  
Topological Descriptors

#### Texture

Spectral Descriptors

#### Moments

#### Principal Components for Description

Principal Components

## Relative positions of pixels:

### Descriptors bases on $G_{norm}$

#### ⑥ Entropy :

$$\sum_{i=1}^K \sum_{j=1}^K p_{ij} \log_2 p_{ij}$$

- Measures the randomness of elements in  $G$
- Entropy = 0: all elements in  $G$  are zeros
- Entropy = 1: all elements in  $G$  are equal
- Max entropy =  $2 \times \log_2 K$



#### Representation

Boundary following  
Chain codes  
Signatures

#### Boundary Descriptors

Basic Descriptors  
Fourier Descriptors

#### Regional Descriptors

Simple Descriptors  
Topological Descriptors

#### Texture

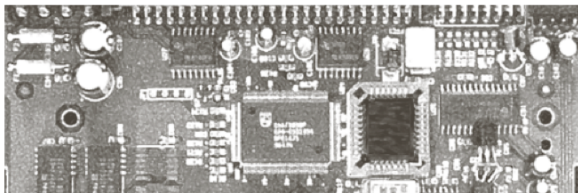
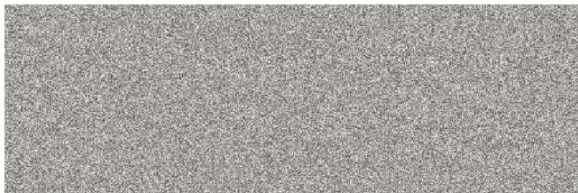
Spectral Descriptors

#### Moments

#### Principal Components for Description

Principal Components

# Texture: Statistical Approaches - Demonstration



a  
b  
c

**FIGURE 11.30**

Images whose pixels have (a) random, (b) periodic, and (c) mixed texture patterns. Each image is of size  $263 \times 800$  pixels.



## Representation

Boundary following  
Chain codes  
Signatures

## Boundary Descriptors

Basic Descriptors  
Fourier Descriptors

## Regional Descriptors

Simple Descriptors  
Topological Descriptors

## Texture

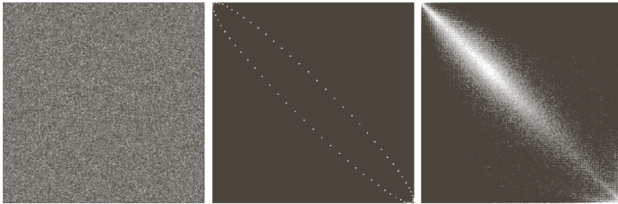
Spectral Descriptors

## Moments

## Principal Components for Description

Principal Components

# Texture: Statistical Approaches - Demonstration



a b c  
**FIGURE 11.31**  
256  $\times$  256 co-occurrence matrices,  $G_1$ ,  $G_2$ , and  $G_3$ , corresponding from left to right to the images in Fig. 11.30.

- $Q$ : One pixel immediately to the right
- Note: these matrices are discriminative



## Representation

Boundary following  
Chain codes  
Signatures

## Boundary Descriptors

Basic Descriptors  
Fourier Descriptors

## Regional Descriptors

Simple Descriptors  
Topological Descriptors

## Texture

Spectral Descriptors

## Moments

## Principal Components for Description

Principal Components

# Texture: Statistical Approaches - Demonstration



## Representation

Boundary following  
Chain codes  
Signatures

## Boundary Descriptors

Basic Descriptors  
Fourier Descriptors

## Regional Descriptors

Simple Descriptors  
Topological Descriptors

## Texture

Spectral Descriptors

## Moments

## Principal Components for Description

Principal Components

| Normalized<br>Co-occurrence<br>Matrix | Descriptor         |             |          |            |             |         |
|---------------------------------------|--------------------|-------------|----------|------------|-------------|---------|
|                                       | Max<br>Probability | Correlation | Contrast | Uniformity | Homogeneity | Entropy |
| $G_1/n_1$                             | 0.00006            | -0.0005     | 10838    | 0.00002    | 0.0366      | 15.75   |
| $G_2/n_2$                             | 0.01500            | 0.9650      | 570      | 0.01230    | 0.0824      | 6.43    |
| $G_3/n_3$                             | 0.06860            | 0.8798      | 1356     | 0.00480    | 0.2048      | 13.58   |

- Measurements are discriminative

# Texture: Statistical Approaches

## Questions

How can you reduce the size of co-occurrence matrices?



### Representation

Boundary following  
Chain codes  
Signatures

### Boundary Descriptors

Basic Descriptors  
Fourier Descriptors

### Regional Descriptors

Simple Descriptors  
Topological Descriptors

### Texture

Spectral Descriptors

### Moments

### Principal Components for Description

Principal Components



## Representation

Boundary following  
Chain codes  
Signatures

## Boundary Descriptors

Basic Descriptors  
Fourier Descriptors

## Regional Descriptors

Simple Descriptors  
Topological Descriptors  
Texture

## Spectral Descriptors

## Moments

## Principal Components for Description

Principal Components

## Method

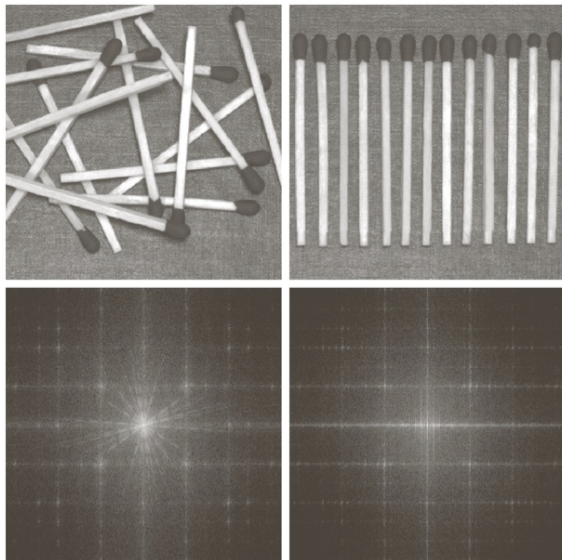
- 1 Compute FFT, called  $F(u.v)$ , for the input image  $f(x, y)$  for sub-image containing the input region.
- 2 Convert  $F(u.v)$  to polar system  $T(r, \theta)$
- 3 Compute measurements  $S(r)$  and  $S(\theta)$

$$S(r) = \sum_{\theta=0}^{\pi} T(r, \theta)$$

$$S(\theta) = \sum_{r=1}^{R_0} T(r, \theta)$$

- $R_0$  : the radius of a circle centered at the origin.

# Spectral Descriptors - Demonstration



|   |   |
|---|---|
| a | b |
| c | d |

**FIGURE 11.35**  
(a) and (b) Images of random and ordered objects.  
(c) and (d) Corresponding Fourier spectra. All images are of size  $600 \times 600$  pixels.



## Representation

- Boundary following
- Chain codes
- Signatures

## Boundary Descriptors

- Basic Descriptors
- Fourier Descriptors

## Regional Descriptors

- Simple Descriptors
- Topological Descriptors
- Texture

## Spectral Descriptors

## Moments

## Principal Components for Description

- Principal Components

# Spectral Descriptors - Demonstration



## Representation

Boundary following  
Chain codes  
Signatures

## Boundary Descriptors

Basic Descriptors  
Fourier Descriptors

## Regional Descriptors

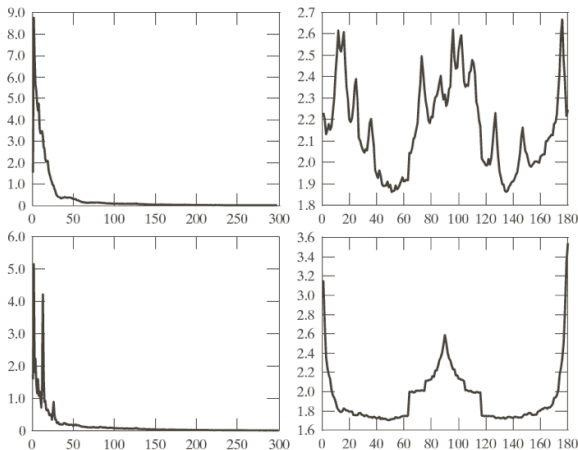
Simple Descriptors  
Topological Descriptors  
Texture

## Spectral Descriptors

## Moments

## Principal Components for Description

Principal Components



a b  
c d

**FIGURE 11.36**  
Plots of (a)  $S(r)$  and (b)  $S(\theta)$  for Fig. 11.35(a). (c) and (d) are plots of  $S(r)$  and  $S(\theta)$  for Fig. 11.35(b). All vertical axes are  $\times 10^5$ .



## Inputs

- 1 Image  $f(x, y)$  of size  $M \times N$

## Moments of order $(p + q)$

$$m_{pq} = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} x^p y^q f(x, y)$$

- $p = 0, 1, 2, \dots$
- $q = 0, 1, 2, \dots$

### Representation

Boundary following  
Chain codes  
Signatures

### Boundary Descriptors

Basic Descriptors  
Fourier Descriptors

### Regional Descriptors

Simple Descriptors  
Topological Descriptors  
Texture  
Spectral Descriptors

### Moments

### Principal Components for Description

Principal Components

## Central Moments of order (p + q)

$$\mu_{pq} = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} (x - \bar{x})^p (y - \bar{y})^q f(x, y)$$

- $p = 0, 1, 2, \dots$
- $q = 0, 1, 2, \dots$

$$\bar{x} = \frac{m_{10}}{m_{00}}$$
$$\bar{y} = \frac{m_{01}}{m_{00}}$$



### Representation

Boundary following  
Chain codes  
Signatures

### Boundary Descriptors

Basic Descriptors  
Fourier Descriptors

### Regional Descriptors

Simple Descriptors  
Topological Descriptors  
Texture  
Spectral Descriptors

### Moments

#### Principal Components for Description

Principal Components

## Set of 7 moments

- See textbook on pp.863
- These moments are invariant to translation, rotation, scaling and mirroring



### Representation

Boundary following  
Chain codes  
Signatures

### Boundary Descriptors

Basic Descriptors  
Fourier Descriptors

### Regional Descriptors

Simple Descriptors  
Topological Descriptors  
Texture  
Spectral Descriptors

### Moments

### Principal Components for Description

Principal Components

# Moments - Demonstration



## Representation

Boundary following

Chain codes

Signatures

## Boundary Descriptors

Basic Descriptors

Fourier Descriptors

## Regional Descriptors

Simple Descriptors

Topological Descriptors

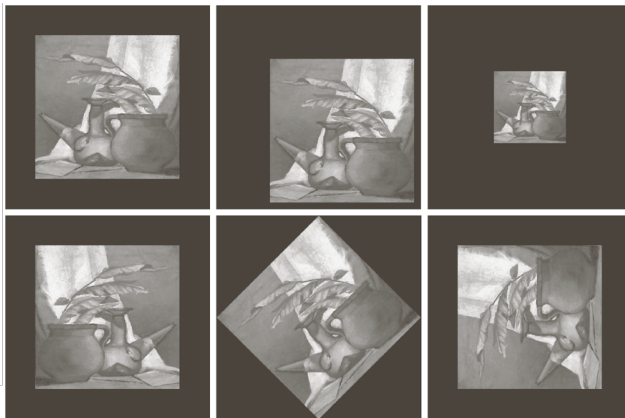
Texture

Spectral Descriptors

## Moments

## Principal Components for Description

Principal Components



**FIGURE 11.37** (a) Original image. (b)–(f) Images translated, scaled by one-half, mirrored, rotated by  $45^\circ$  and rotated by  $90^\circ$ , respectively.

# Moments - Demonstration



## Representation

Boundary following  
Chain codes  
Signatures

## Boundary Descriptors

Basic Descriptors  
Fourier Descriptors

## Regional Descriptors

Simple Descriptors  
Topological Descriptors  
Texture  
Spectral Descriptors

## Moments

## Principal Components for Description

Principal Components

| Moment Invariant | Original Image | Translated | Half Size | Mirrored | Rotated 45° | Rotated 90° |
|------------------|----------------|------------|-----------|----------|-------------|-------------|
| $\phi_1$         | 2.8662         | 2.8662     | 2.8664    | 2.8662   | 2.8661      | 2.8662      |
| $\phi_2$         | 7.1265         | 7.1265     | 7.1257    | 7.1265   | 7.1266      | 7.1265      |
| $\phi_3$         | 10.4109        | 10.4109    | 10.4047   | 10.4109  | 10.4115     | 10.4109     |
| $\phi_4$         | 10.3742        | 10.3742    | 10.3719   | 10.3742  | 10.3742     | 10.3742     |
| $\phi_5$         | 21.3674        | 21.3674    | 21.3924   | 21.3674  | 21.3663     | 21.3674     |
| $\phi_6$         | 13.9417        | 13.9417    | 13.9383   | 13.9417  | 13.9417     | 13.9417     |
| $\phi_7$         | -20.7809       | -20.7809   | -20.7724  | 20.7809  | -20.7813    | -20.7809    |

**TABLE 11.5**  
Moment invariants for the images in Fig. 11.37.

- Note: The value of moments just **change slightly** with rotation, scalling, translation, and mirroring.



## Inputs

- ① A set  $X$  of  $K$  vectors, each has size of  $n \times 1$

$$\mathbf{x}_k = \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ x_n \end{bmatrix}$$

- $\mathbf{x}$  : an **observation** (measurement) for  $n$  **variables** (features).
- The variables can be width, height, length, or any other extracted features.
- $\mathbf{x}$ : can be thought as a **point in n-dimension space**.

### Representation

Boundary following  
Chain codes  
Signatures

### Boundary Descriptors

Basic Descriptors  
Fourier Descriptors

### Regional Descriptors

Simple Descriptors  
Topological Descriptors  
Texture  
Spectral Descriptors

### Moments

### Principal Components for Description

Principal Components



## Basic Ideas

- Components or variables in vectors of  $X$  are correlated or uncorrelated.
- Principal Component Analysis (PCA) will propose a **transforms**. When we apply this transforms for each vector in  $X$ , we obtain a new set of vectors that their variables (or components) are **uncorrelated**.

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## Steps

- 1 Compute the mean vector of  $X$

$$\begin{aligned}\mathbf{m}_x &= E\{\mathbf{x}\} \\ &\approx \frac{1}{K} \sum_{k=1}^K \mathbf{x}_k\end{aligned}$$

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## Steps

- ② Compute the variance-covariance matrix (called covariance matrix) of  $X$

$$\begin{aligned}\mathbf{C}_x &= E\{(\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^T\} \\ &\approx \frac{1}{K} \sum_{k=1}^K \mathbf{x}_k \mathbf{x}_k^T - \mathbf{m}_x \mathbf{m}_x^T\end{aligned}$$

- $\mathbf{C}_x$ : real and symmetric matrix

## Meaning of the mean vector and the covariance matrix

- **Mean vector:**
  - A vector that contains the mean value for each component (variable)
  - The centroid of the shape, if vectors are points in space.
- **Covariance matrix:**
  - Elements on diagonal: The variance of components
  - Elements off-diagonal: The covariance between component  $x$  and component  $y$ . These covariances are either negative or positive.
    - Positive covariance: Large value on component  $x$  tends to occur with large value on component  $y$ . Small value on component  $x$  tends to occur with small value on component  $y$
    - Negative covariance: Large value on component  $x$  tends to occur with small value on component  $y$  and vice versa.



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## Steps

③ Calculate eigenvectors and eigenvalues of  $\mathbf{C}_x$

Let  $\mathbf{e}_i$  and  $\lambda_i$  be the eigenvectors and corresponding eigenvalues of  $\mathbf{C}_x$ , for  $i = 1, 2, \dots, n$

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# Principal Components



## Steps

- ④ Form transforms matrix **A** as follows:
- Matrix **A** has size of  $n \times n$
  - Eigenvectors are put to **rows** of **A** sin such a way that their corresponding eigenvalues are in descending-order from the first row to the last row in matrix **A**

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## Transforms Matrix's Properties

- **A** is an orthogonal matrix, so
- $\mathbf{A} = \mathbf{A}^T$



**A** is used to transform input vectors in  $X$  to principal components

## Steps

- 5 Perform the transformation

$$\mathbf{y} = \mathbf{A}(\mathbf{x} - \mathbf{m}_x)$$

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### Properties of set of vector $\mathbf{y}$

- ① The mean vector of  $\mathbf{y}$  is vector  $\mathbf{0}$

$$\begin{aligned}\mathbf{m}_y &= E\{\mathbf{y}\} \\ &= \mathbf{0}\end{aligned}$$

- ② Variables of  $\mathbf{y}$  are uncorrelated  $\equiv$  Covariance matrix of vectors  $\mathbf{y}$  has all elements zeros, except ones on diagonal.

$$\begin{aligned}\mathbf{C}_y &= \mathbf{A}\mathbf{C}_x\mathbf{A}^T \\ &= \begin{bmatrix} \lambda_1 & 0 & 0 & .. & 0 \\ 0 & \lambda_2 & 0 & .. & 0 \\ 0 & 0 & \lambda_3 & .. & 0 \\ 0 & .. & 0 & \lambda_n & 0 \end{bmatrix}\end{aligned}$$

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Reconstruction  $\mathbf{x}$  from  $\mathbf{y}$  - **perfect reconstruction**

$$\mathbf{x} = \mathbf{A}^T \mathbf{y} + \mathbf{m}_x$$

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# Principal Components

- $\mathbf{A}_k$ : created from  $\mathbf{A}$  by keeping the first  $k$  rows
  - $\Rightarrow \mathbf{A}_k$  has size:  $k \times n$
  - $\Rightarrow \mathbf{A}_k^T$  has size:  $n \times k$

## Reconstruction $\mathbf{x}$ from $\mathbf{y}$ - Approximation

- 1 Keep  $k$  important components of  $\mathbf{y}$ :

$$\mathbf{y} = \mathbf{A}_k(\mathbf{x} - \mathbf{m}_x)$$

- 2 Reconstruct  $\mathbf{x}$  from  $\mathbf{y}$ :

$$\hat{\mathbf{x}} = \mathbf{A}_k^T \mathbf{y} + \mathbf{m}_x$$

- $\Rightarrow \mathbf{y}$  has size:  $k \times 1$
- $\Rightarrow \hat{\mathbf{x}}$  has size:  $n \times 1$
- $\hat{\mathbf{x}}$  is an approximation of  $\mathbf{x}$  using the first  $k$  important components.





## Dimension reduction

$$\mathbf{y} = \mathbf{A}_k(\mathbf{x} - \mathbf{m}_x)$$

- ① The first  $k$  important component can be used as descriptor, so
- ② Number of dimensions are reduced from  $n$  down to  $k$ 
  - Size of input  $\mathbf{x}$ :  $n$
  - Size of output  $\mathbf{y}$ :  $k$

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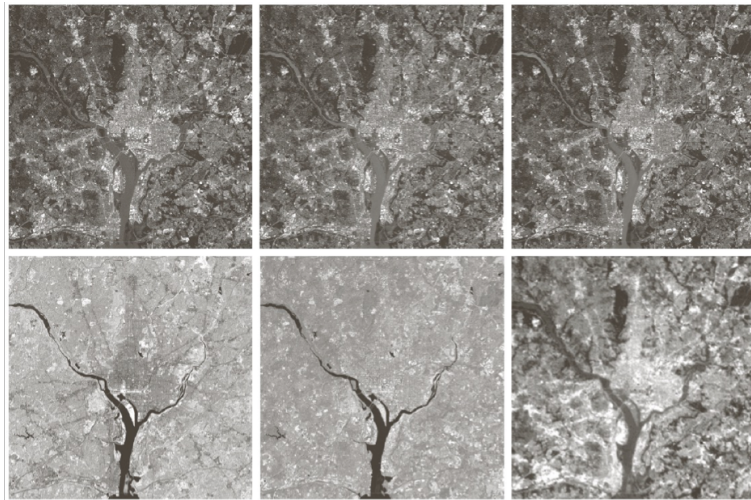
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# Principal Components: Applications



a b c  
d e f

**FIGURE 11.38** Multispectral images in the (a) visible blue, (b) visible green, (c) visible red, (d) near infrared, (e) middle infrared, and (f) thermal infrared bands. (Images courtesy of NASA.)

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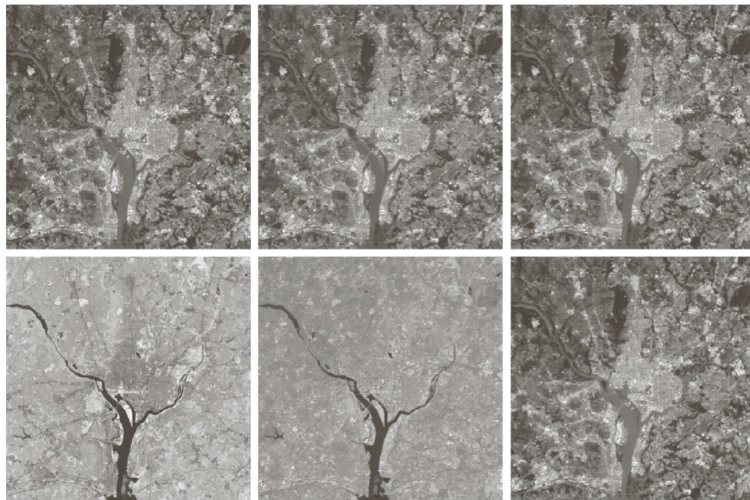
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# Principal Components: Applications



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**FIGURE 11.41** Multispectral images reconstructed using only the two principal component images corresponding to the two principal component images with the largest eigenvalues (variance). Compare these images with the originals in Fig. 11.38.

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$$\mathbf{y} = \mathbf{A}(\mathbf{x} - \mathbf{m}_x)$$

## Properties of $\mathbf{y}$

Transformed features  $\mathbf{y}$  has the following advantages compared to  $\mathbf{x}$ :

- ① Invariant to translation and rotation
  - $\mathbf{y}$  has been shifted to the centroid by  $\mathbf{m}_x$
  - $\mathbf{y}$  has been aligned with principal directions (eigenvectors)
- ② Invariant to scaling can be achieved by dividing  $\mathbf{y}$  for eigenvalues.

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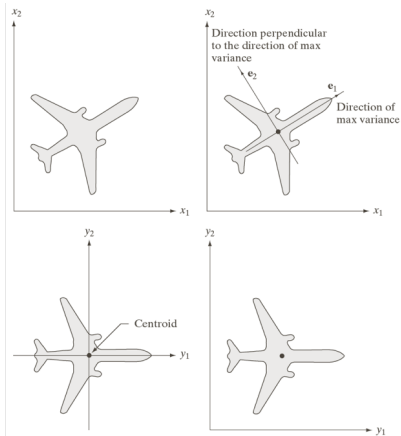
# Principal Components: Applications

$$\mathbf{m}_x = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

$$\mathbf{C}_x = \begin{bmatrix} 3.333 & 2.00 \\ 2.00 & 3.333 \end{bmatrix}$$

$$\mathbf{e}_1 = \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix}$$

$$\mathbf{e}_2 = \begin{bmatrix} -0.707 \\ 0.707 \end{bmatrix}$$



a b  
c d

**FIGURE 11.43**

(a) An object.  
(b) Object showing eigenvectors of its covariance matrix.  
(c) Transformed object, obtained using Eq. (11.4-6).  
(d) Object translated so that all its coordinate values are greater than 0.



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